JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY GURAJADA VIZIANAGARAM

SRI SIVANI COLLEGE OF ENGINEERING (AFFILIATED TO JNTU-GV)

DEPARTMENT OF ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

★ OFFICIAL ACADEMIC SYLLABUS & BLUEPRINT ★

B.Tech AI & ML R23 Mathematical Foundations & Python for Data Science Syllabus

Course Code:	R23-AIML-1102	**Academic Scheme:**	I B.Tech I Semester (R23 Regulation)
Verification:	Verified Authentic Academic Copy	**System Ref:**	STUDY-BLUEPRINT-VERIFIED-2026

DETAILED CURRICULAR MODULES & TOPICS

Unit I: Linear Algebra Foundations for Machine Learning

Vectors, Norms, Orthogonality, Gram-Schmidt process, Singular Value Decomposition (SVD), Principal Component Analysis (PCA) derivation, Matrix calculus.

Unit II: Probability, Random Variables & Distributions

Conditional probability, Bayes Theorem, Random variables, Expectation, Variance, Discrete and continuous distributions (Binomial, Poisson, Normal, Gaussian), Central Limit Theorem.

Unit III: Python Programming & Scientific Libraries

Python syntax, Data structures (Lists, Tuples, Dictionaries, Sets), NumPy n-dimensional arrays, vectorization, broadcasting, indexing and slicing.

Unit IV: Data Wrangling & Visualization with Pandas

Series, DataFrames, Missing value imputation, GroupBy, Merging and joins, Matplotlib plotting, Seaborn statistical charts, Heatmaps.

Unit V: Exploratory Data Analysis & Feature Engineering

Outlier detection using IQR and Z-scores, Standard scaling and Min-Max normalization, One-Hot encoding, Feature selection using correlation matrices.

EVALUATION SCHEME & EXAMINATION BLUEPRINT

Internal Evaluation	30 Marks
Semester End Examination	70 Marks
Branch Code	AIML / CSE(AI&ML;)

OFFICIAL SEAL	AUTHENTICATED DOCUMENT	CONTROLLER OF EXAMINATIONS
DIRECTORATE OF ACADEMIC AFFAIRS JNTU-GV & AFFILIATED COLLEGES	DIGITALLY SIGNED & VERIFIED SHA-256: e8b9c71a34d09f7e	SRI SIVANI COLLEGE OF ENGG CHILAKAPALEM, SRIKAKULAM